

One Policy, Many Vehicles

Teacher–student distillation for 6-DOF underwater vehicles

Arne Jacobs

H00508832

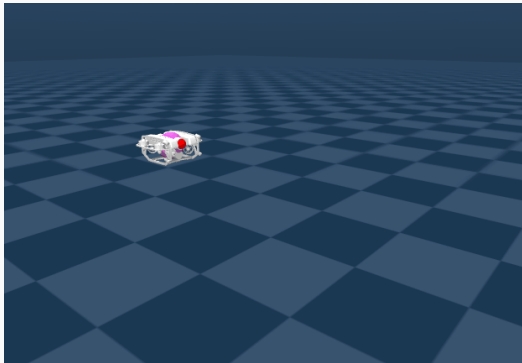
August 10, 2026

MSc Robotics — MSc Project Presentation

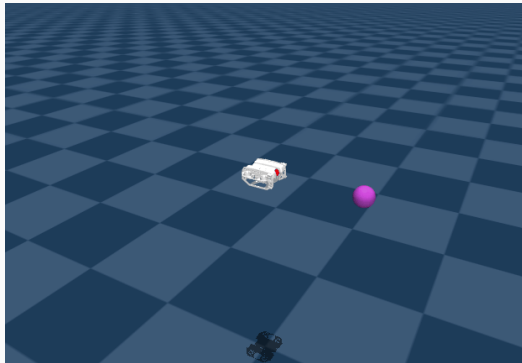
Heriot-Watt University — Ocean Systems Laboratory

Supervised by Dr Ignacio Carlucho

What this project is



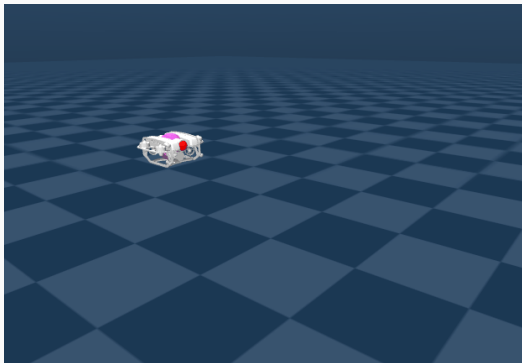
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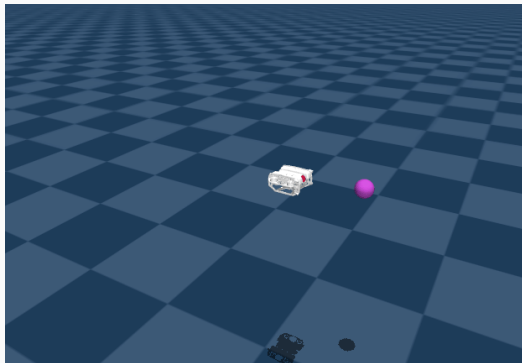
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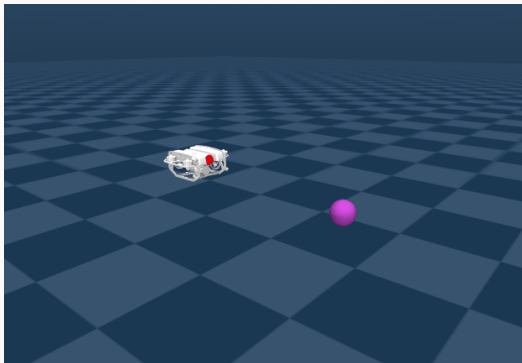
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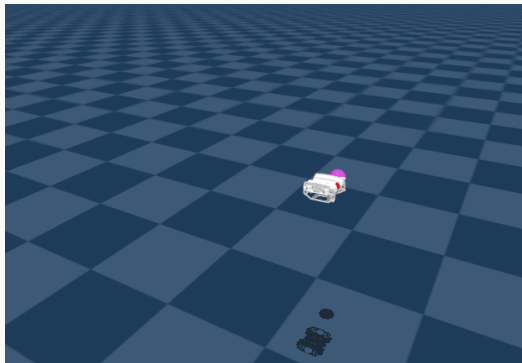
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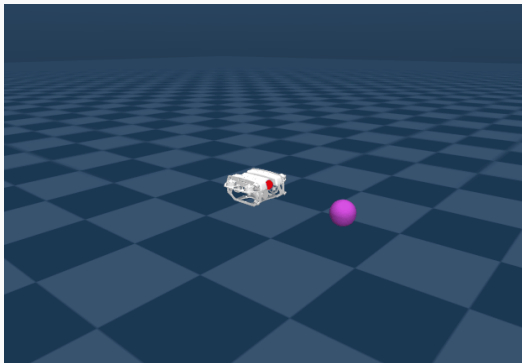
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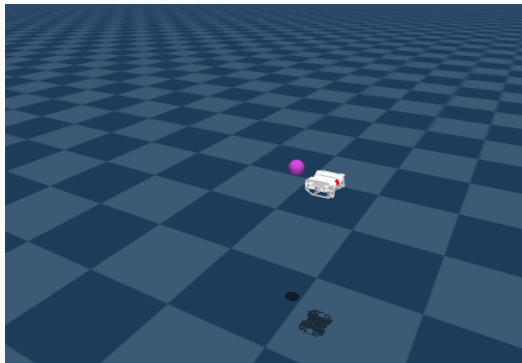
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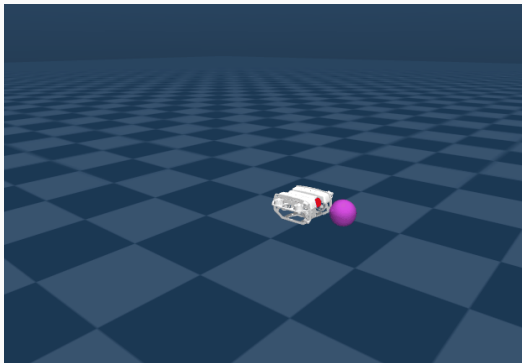
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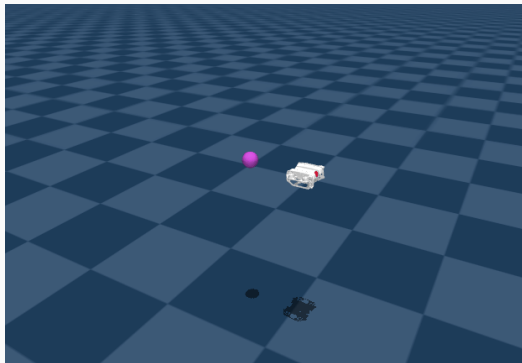
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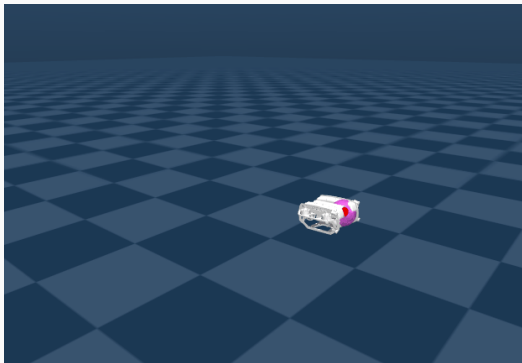
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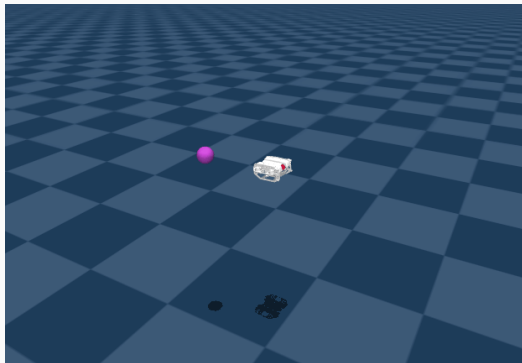
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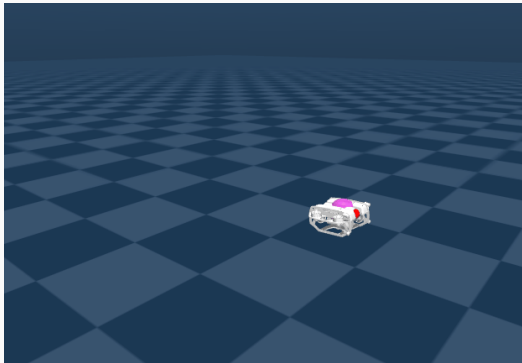
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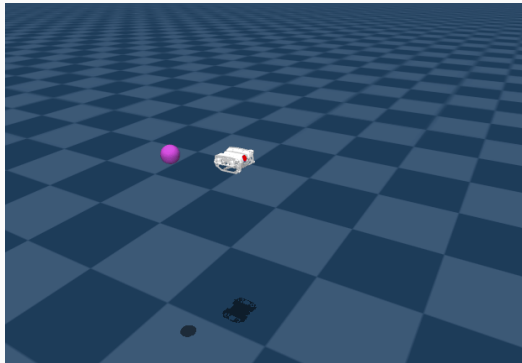
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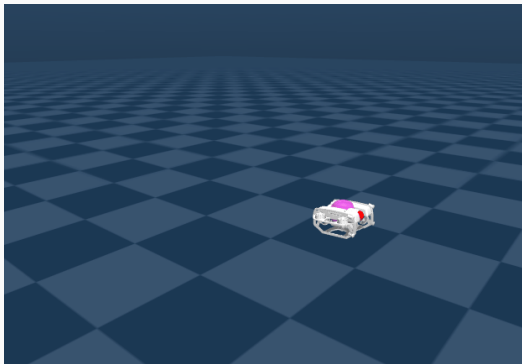
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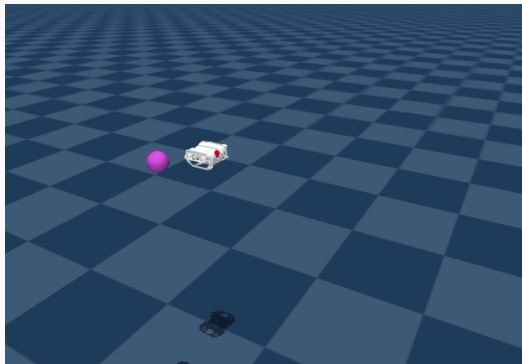
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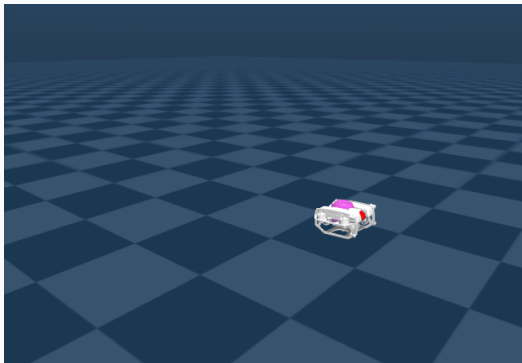
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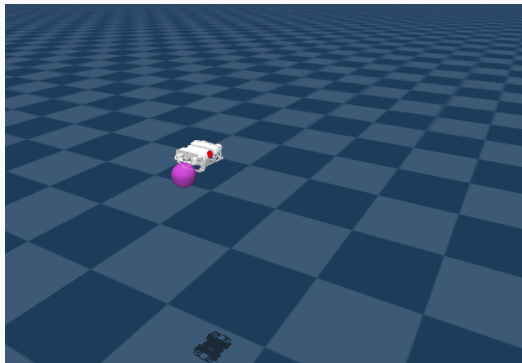
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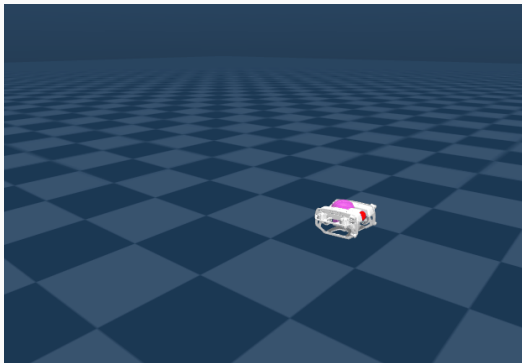
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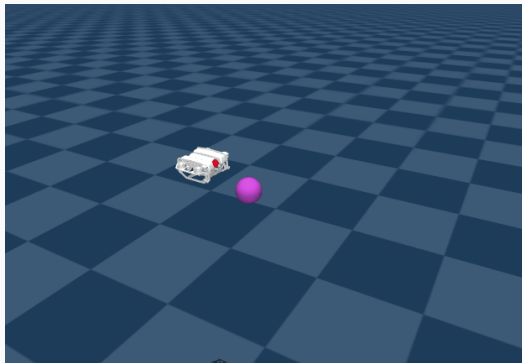
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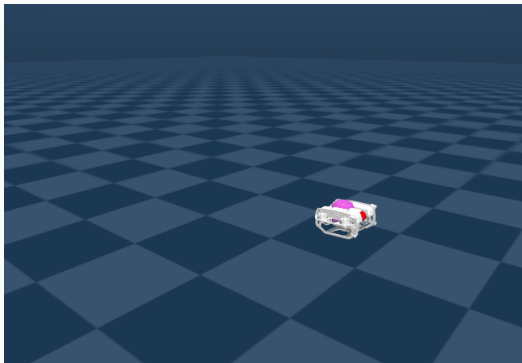
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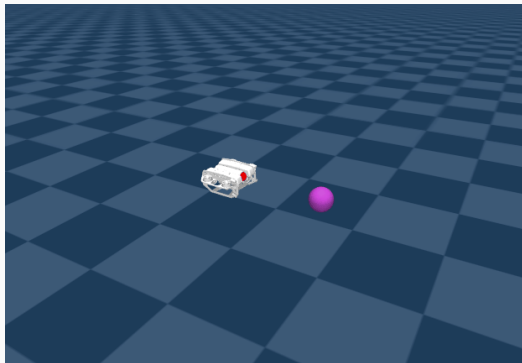
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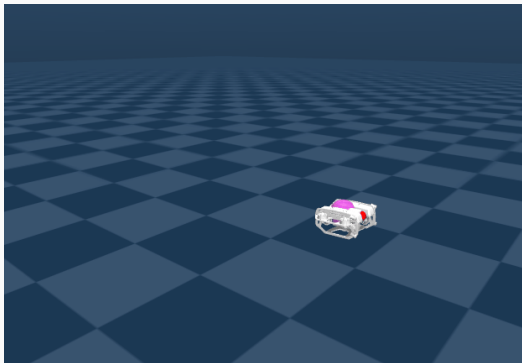
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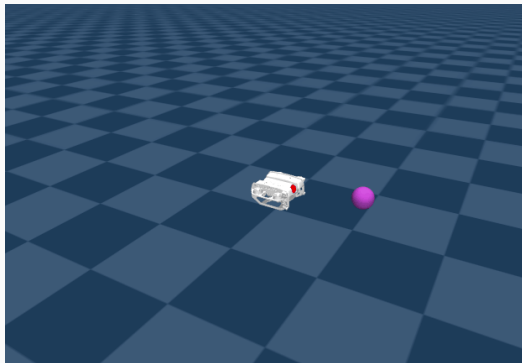
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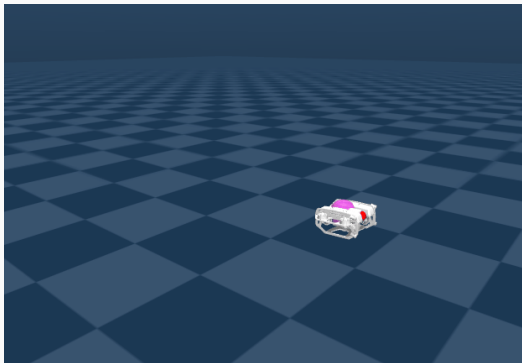
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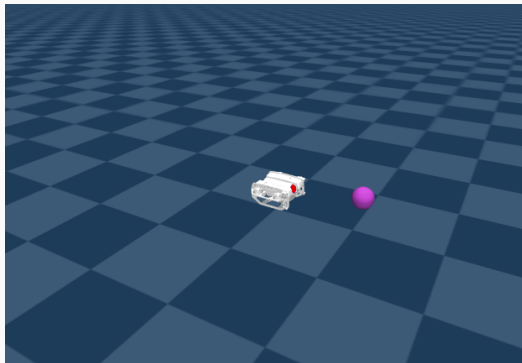
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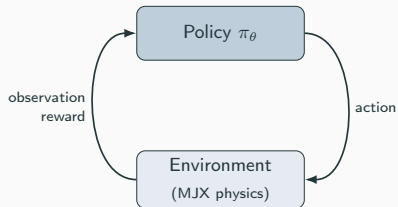
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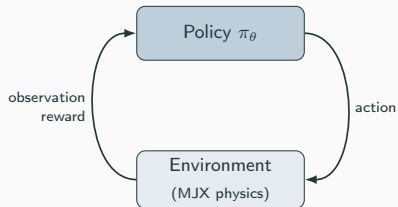
Background: four ideas the rest of the talk rests on



Reinforcement learning

Learn π_θ by **trial and error** against a reward,
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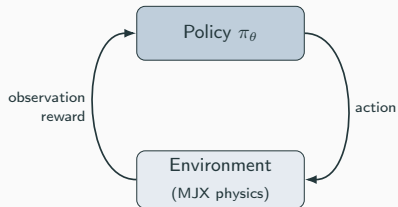


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PPO $\Rightarrow$ Teacher. On-policy. Clipped surrogate objective.
Small updates making sure 1,000 unattended runs converge.

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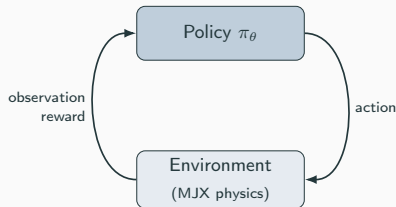
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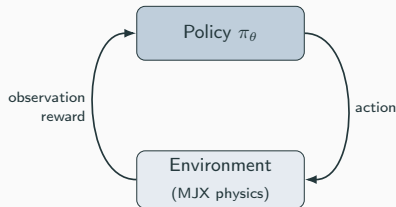
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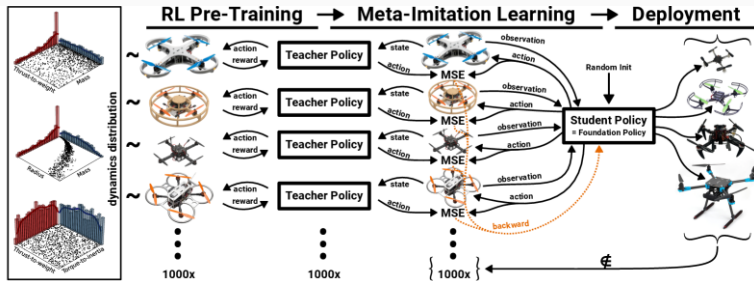
Imitation / distillation $\Rightarrow$ Student. Supervised learning. **Dagger**: the student drives and labels its own states based on the teacher's actions.

Recurrence (GRU) $\Rightarrow$ Adaptation. Recurrent policy. Carries a **hidden state** across timesteps, so its action can depend on its history.

Two projects, combined

RAPTOR

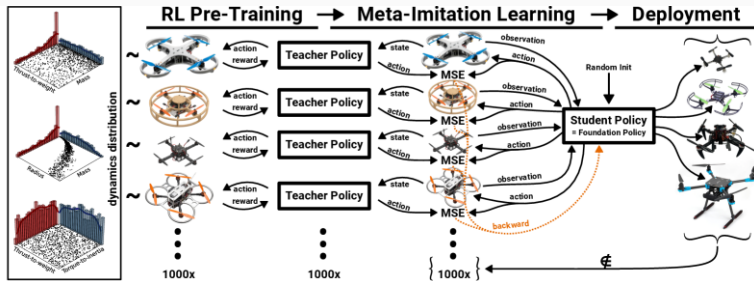
Eschmann et al. [1]



Two projects, combined

RAPTOR

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Fast6DOF

Tunçay et al. [2]

Ocean Systems Laboratory

GPU-parallel 6-DOF AUV control in JAX + MJX.

A policy in **minutes**, zero-shot to the real vehicle.

I inherited: the physics, the reward, and the action encoding.

Approach: population of specialists → one generalist

AUV bank

100 bodies × 10 buoyancy draws

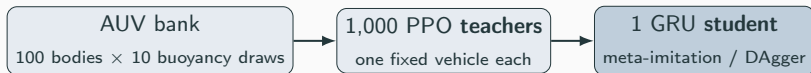
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- Gated Recurrent Unit (GRU)
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Research questions

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- RQ4** How does it compare against a **matched randomisation baseline**?

Origin of the study

The group's prior experience was that buoyancy randomisation *does not work*.

Thus I investigated the RAPTOR style pipeline.

I eventually tested the buoyancy randomisation to use it as a baseline, leading to RQ4.

Objectives set before starting

- settled error < 0.05 in both channels
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Four independent instruments

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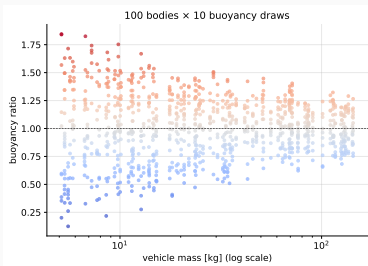
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Four independent instruments

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I report *settled* error i.e. the last 40% of each episode, not the flight to the target.

The AUV bank, and the 1,000 teachers trained on it



How a vehicle is sampled:

Mass **log-uniform 5–150 kg**: equal coverage per decade

Other size, inertia, thrust, torque from *mass-scaling laws*
($\ell \propto m^{1/3}$, $I \propto m^{5/3}$, $T \propto m^{2/3}$)

Jitter per-axis, so equal-mass hulls still differ

Buoyancy drawn as a fraction of *the vehicle's* span

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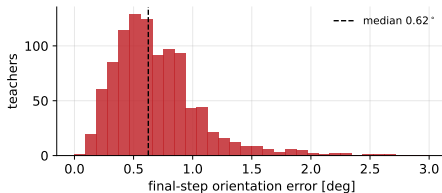
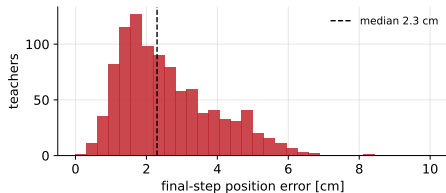
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Buoyancy drawn as a fraction of *the vehicle's span*

Every one of the 1000 teachers converged



Then distillation hit a wall, and three fixes did nothing

Building the first student Arm A: buoyancy-diverse teachers

Intervention	Loss	Pos. err.
<i>Non-buoyancy reference</i>	<i>0.015</i>	<i>0.34 cm</i>
Baseline (re-randomised buoyancy)	0.045	4.18 cm
Capacity 16 $\rightarrow$ 64 ($\times 16$ params)	0.046	4.14 cm
+ integral-error observation	0.050	4.43 cm
+ gravity-direction observation	0.052	4.46 cm
Averaged across every buoyancy draw in the bank		

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Label-consistency fix	0.020	1.91 cm

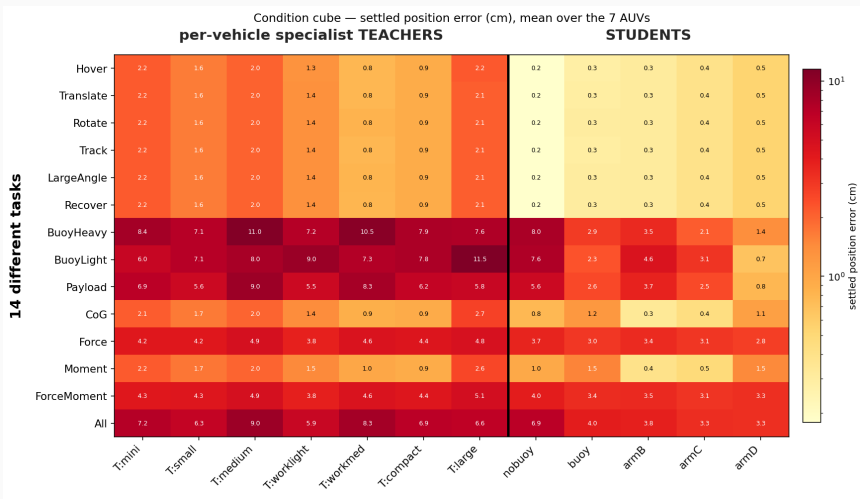
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Root cause

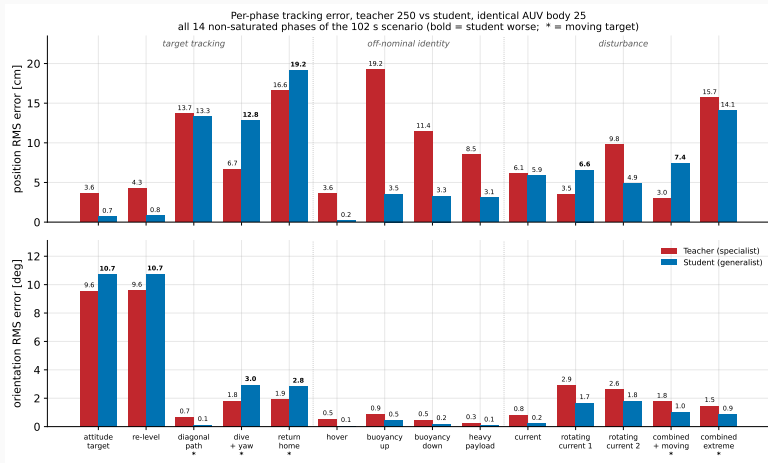
Buoyancy was re-sampled every epoch during student rollout.

DEMO

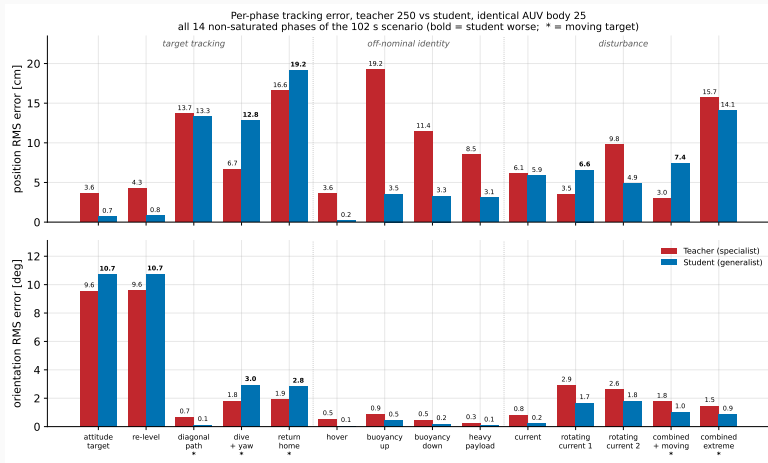
RQ1: the generalist beats the specialists off-nominal



Contrary evidence: where the student loses

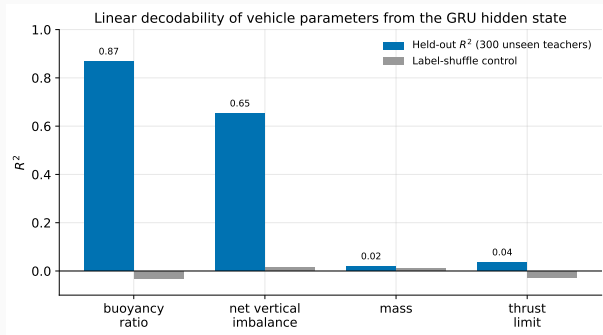


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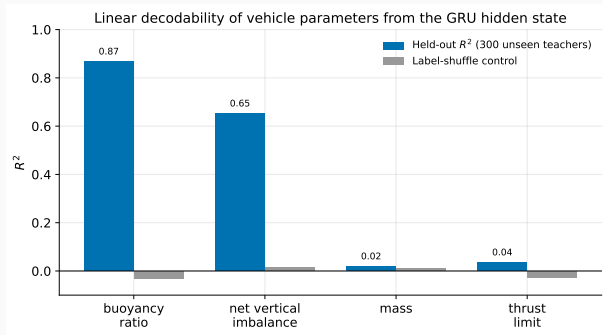
A trade, not a clean sweep $\Rightarrow$ robustness to the *vehicle* costs sharpness on a *moving* reference.

RQ2: it really does estimate the vehicle



A linear probe reads the **buoyancy** of a *held-out* vehicle straight out of the 16-unit hidden state.

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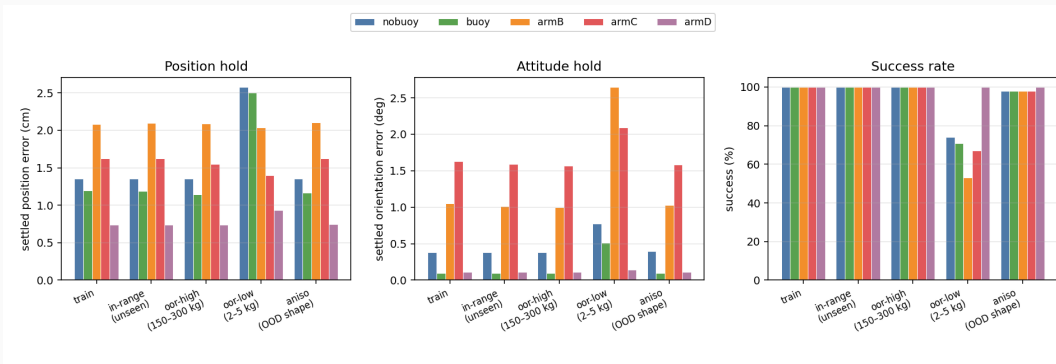


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Mass and thrust decode at almost **zero**.

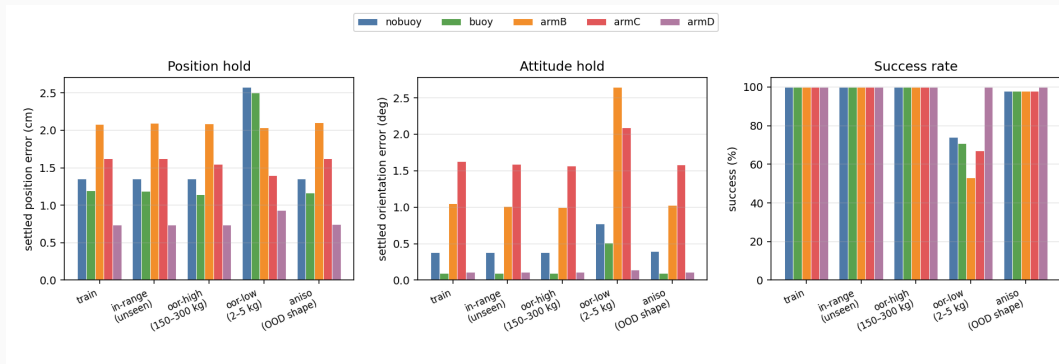
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RQ3a: generalisation to vehicles never seen



Four fresh banks no policy ever trained on — seed 999, disjoint from the training bank

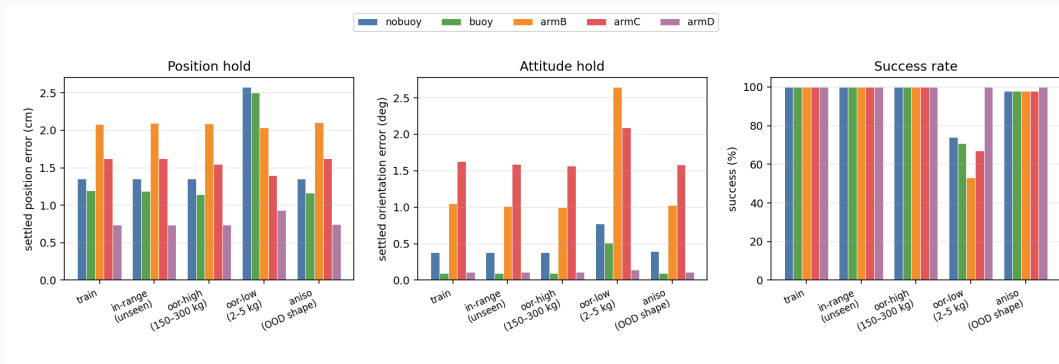
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Flat across 100 unseen in-range vehicles, **2×** the maximum training mass, and **5:1** anisotropic hulls.

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Flat across 100 unseen in-range vehicles, **2×** the maximum training mass, and **5:1** anisotropic hulls.

Only limit: the **light-mass boundary** (2–5 kg), where success drops to **53–71%**.

RQ3b: sim-to-sim: a different engine, a different vehicle



BlueROV2-Heavy in Stonefish.

The orange lines are thruster force vectors.

Stonefish + BlueROV2-Heavy, all five students on the *identical* trajectory. Settled hover:

Student	hover	depth swing
buoyancy	6.20 cm	11.9 cm
CoG	4.48 cm	12.0 cm
CoG + delay	3.08 cm	8.9 cm
delay	1.03 cm	2.4 cm

Real actuation lag: Stonefish: ≈ 50 ms vs MJX **none**.

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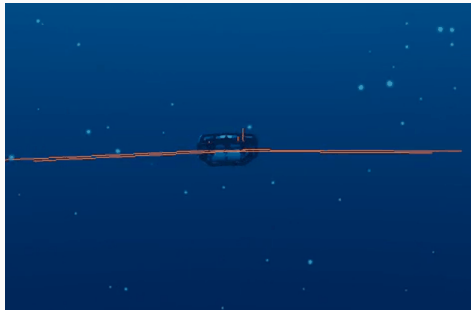
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635 seconds $\approx$ one teacher-training, against a thousand plus distillation

$\Rightarrow$ **10.5 minutes** vs **6.5 days!**

- Specialist: sharp **V**
- Student: shallow bowl
- **Baseline: a very similar bowl**

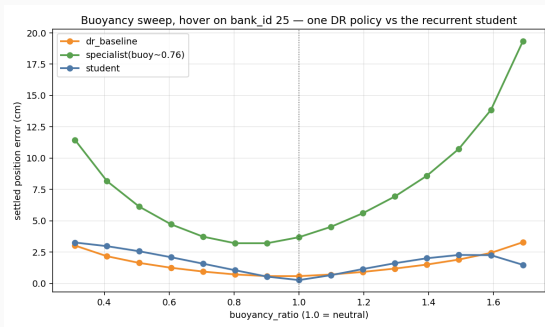
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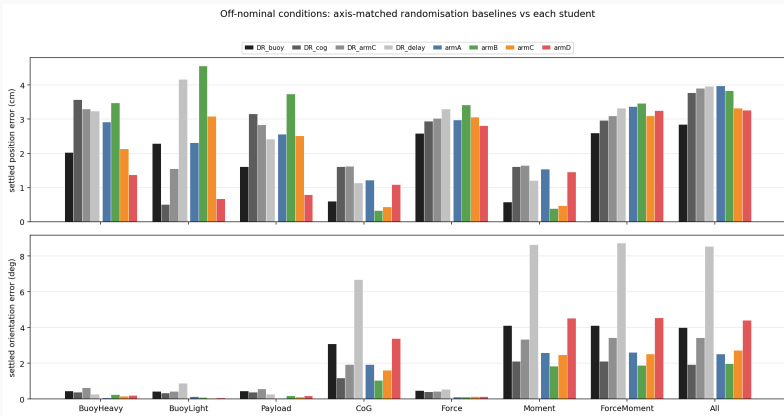
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So I ran the missing control

The baseline randomised **one** axis; the students were distilled over **several**.

⇒ **three more baselines**, one per arm, **minutes each**.



Position splits 4–4 & attitude 7–8 for the student

| DR_delay collapses, 8.5° against armD's 4.4°

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What recurrence is actually for

Composing several identity axes without paying for each.
Not better adaptation along any one.
Cost: graceful degradation outside the training range.

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Where it goes next

Continuing with **Sümer Tunçay** toward a joint paper
⇒ target **IEEE OCEANS**, in Scotland next year

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- Drag never randomised; the heavy end validated in **one engine only**

Thank you

Backup: the centre-of-gravity design bug

Sampling the CoG offset as a flat fraction of *geometry* ignores torque authority. Required righting torque $\sim mgd$ grows with mass:

demand/available torque: $0.75 \rightarrow 2.11$ (mini $\rightarrow$ large).

Above 1.0 the vehicle **cannot right itself**.

Heaviest class: 40 cm, 47° , **0% success** — uncontrollable by construction.

Fix: cap the offset to a fraction of each vehicle's own torque authority.

Result: every class ≈ 3 cm and 1.1 – 1.4° ; expert rate **33% $\rightarrow$ 90.3%**.

Same authority-bounding lesson the buoyancy limit taught in the translational channel.

Backup: actuation delay makes the problem non-Markovian

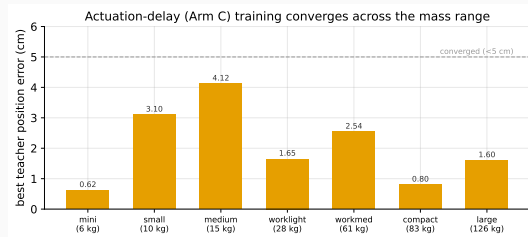
A first-order actuator lag means two states with identical error and velocity evolve *differently* depending on the in-flight wrench.

A memoryless teacher cannot see that wrench
⇒ cannot emit a consistent action. Pilot stalled at ≈ 0.5 m for every class.

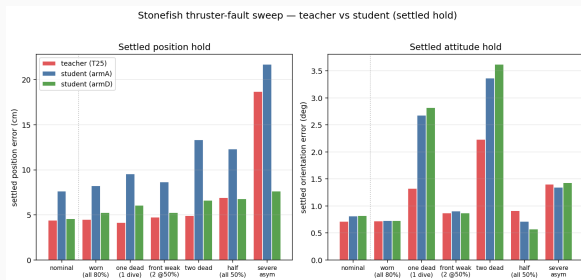
Fix: state augmentation — append the 6-d lagged wrench to the observation (12-d → 18-d).

Every class then converged: 0.6–4.1 cm.

The fix came from control theory, not from a learning-side adjustment.



Backup: thruster-fault sweep (out-of-distribution actuators)



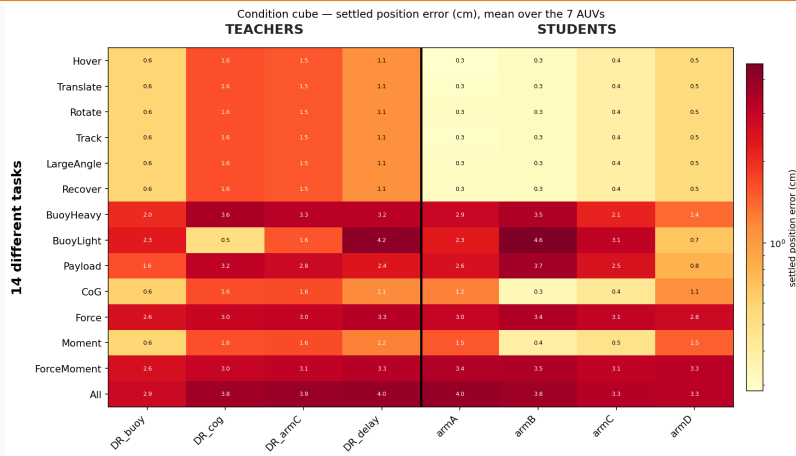
Seven configurations, mild wear → severe asymmetry → dead units. No policy trained for any of them.

Severe asymmetry: Arm D **7.6 cm** vs Arm A 21.7 cm vs specialist 18.7 cm.

Caveat I must state: the randomisation baselines were **not run in this sweep**, so this compares students against a specialist only. It is the one comparison the matched baselines did *not* repair, and the only architectural claim I do not make.

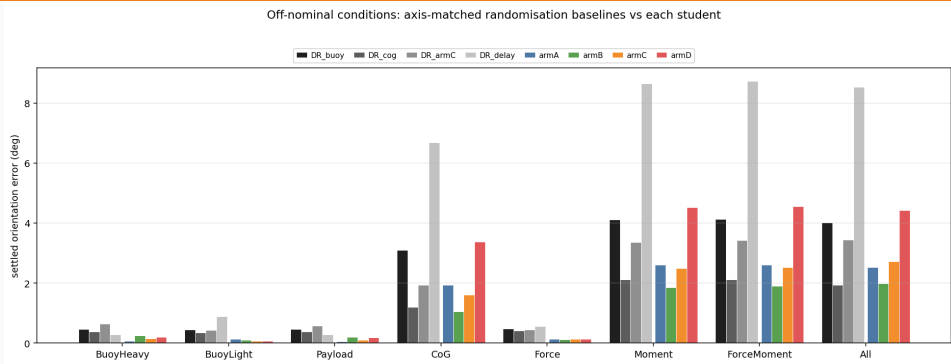
Methodological note: Arm D's action trace looks saturated and chattering — that is the delay-trained high-gain law, not instability. Read settled hold, not the action trace.

Backup: condition cube — DR baselines vs students



Each baseline is matched to the student beside it, axis for axis. The flat top block is the nominal conditions — the baselines carry a raised, condition-invariant floor, their own hover precision rather than the stressor.

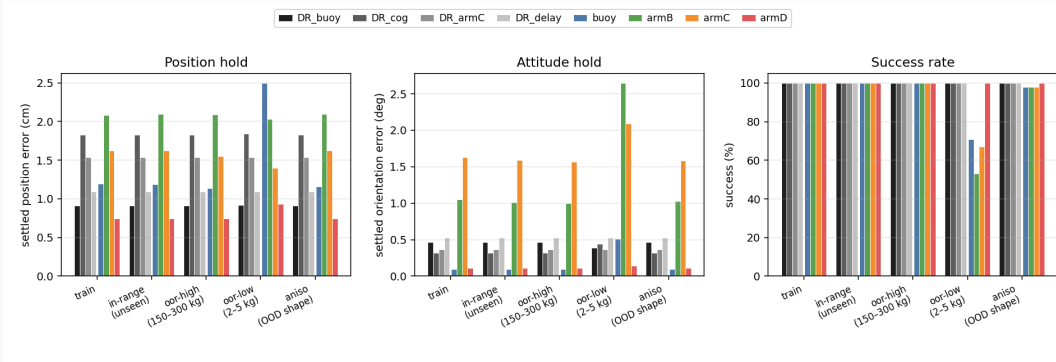
Backup: the matched comparison in attitude



DR_delay is the outlier — fine at nominal (0.52°) and *better* than DR_buoy on force-only conditions (0.29 vs 0.47°), but it loses attitude entirely wherever a **sustained torque** is applied: 6.7° under CoG offset, 8.7° under an applied moment.

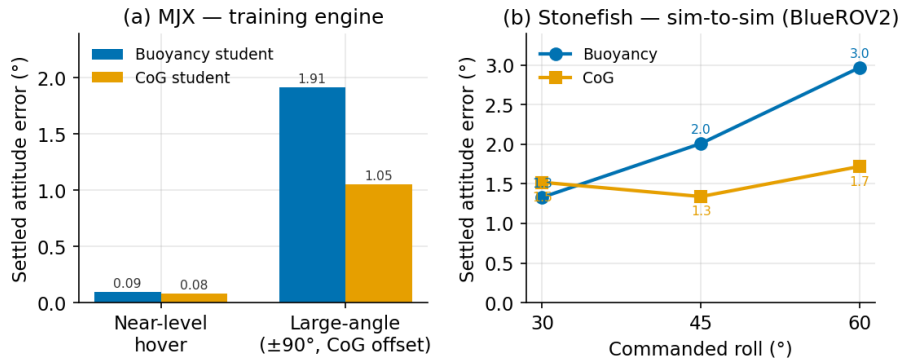
Delay training yields a **high-gain** law, and that arm never trained against a persistent moment. Corroboration: armD (same axes, recurrent) shows the same signature halved; DR_armC (delay *plus* CoG training) is rescued to 1.9 . So it is not delay — it is **delay without attitude-disturbance training**.

Backup: held-out — where the baselines win



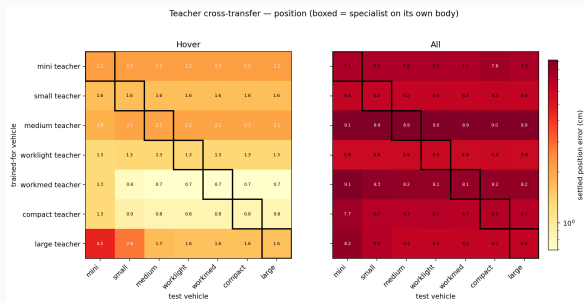
All four baselines hold **100%** success on every bank. At the 2–5 kg boundary armB drops to **53%**, armC **67%**, armA **71%** — only armD matches. The students *infer* the vehicle, and an inference outside its training support returns a confident wrong answer; a policy that never learned a body-dependent mapping has nothing to extrapolate wrongly.

Backup: attitude across both engines



The centre-of-gravity arm holds tilt more tightly than the buoyancy arm in *both* engines — 1.05° vs 1.91° at large angle in MJX, and 1.7° vs 3.0° at 60° commanded roll in Stonefish. The design choice survives the engine change, which is the part worth claiming; the absolute numbers do not transfer and are not claimed.

Backup: cross-transfer — why body scale is free



The **rows are flat**: a 5 kg specialist scores the same on a 143 kg hull as on its own. The boxed diagonal is not special.

Because the action is normalised to each vehicle's own authority, and that authority scales with mass, the closed-loop response is approximately mass-invariant *by construction*.

Consequence: the $28\times$ mass range — the most impressive-looking axis — is not where the difficulty lies. It also means the baseline's single-body training is a much smaller handicap than it appears.

Backup: numbers at a glance

	Nominal pos.	Nominal ori.	All-stressors	Held-out
Specialist teachers	2.0–2.2 cm	—	—	—
DR baseline	0.60 cm	0.46°	2.85 cm	0.91 cm
Arm A (buoyancy)	0.28 cm	0.10°	4.00 cm	1.19 cm
Arm B (CoG)	0.31 cm	0.08°	3.80 cm	2.08 cm
Arm C (CoG+delay)	0.39 cm	0.08°	3.30 cm	1.63 cm
Arm D (delay)	0.54 cm	0.11°	3.30 cm	0.74 cm

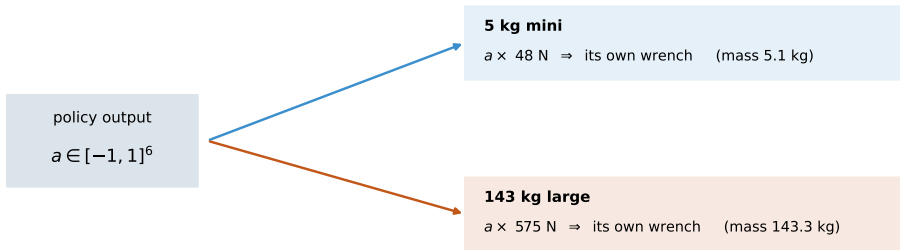
Settled error, mean over seven vehicles spanning 5–150 kg. Held-out = mean over four unseen banks.

What Fast6DOF already solved: the action encoding



The policy commands a FRACTION of available authority, never an absolute force.
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Body scale is already free. What is NOT free is buoyancy — and that is the project.



Jonas Eschmann, Dario Albani, and Giuseppe Loianno.

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